3.5 Fundamentals of data representation

3.5.1 Number systems

3.5.1.1 Natural numbers

Content	Additional information
Be familiar with the concept of a natural number and the set $\mathbb{N}$ of natural numbers (including zero).	$\mathbb{N} = \{0, 1, 2, 3, \ldots\}$

3.5.1.2 Integer numbers

Content	Additional information
Be familiar with the concept of an integer and the set $\mathbb{Z}$ of integers.	$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$

3.5.1.3 Rational numbers

Content	Additional information
Be familiar with the concept of a rational number and the set $\mathbb{Q}$ of rational numbers, and that this set includes the integers.	$\mathbb{Q}$ is the set of numbers that can be written as fractions (ratios of integers). Since a number such as 7 can be written as 7/1, all integers are rational numbers.

3.5.1.4 Irrational numbers

Content	Additional information
Be familiar with the concept of an irrational number.	An irrational number is one that cannot be written as a fraction, for example $\sqrt{2}$.

3.5.1.5 Real numbers

Content	Additional information
Be familiar with the concept of a real number and the set $\mathbb{R}$ of real numbers, which includes the natural numbers, the rational numbers, and the irrational numbers.	$\mathbb{R}$ is the set of all 'possible real world quantities'.

3.5.1.6 Ordinal numbers

Content	Additional information
Be familiar with the concept of ordinal numbers and their use to describe the numerical positions of objects.	When objects are placed in order, ordinal numbers are used to tell their position. For example, if we have a well-ordered set S = {'a', 'b', 'c', 'd'}, then 'a' is the 1st object, 'b' the 2nd, and so on.

3.5.1.7 Counting and measurement

Content	Additional information
Be familiar with the use of: • natural numbers for counting • real numbers for measurement.	